

Unit 2 – Exploring Two Variable Data
HW 2 – Scatterplots and Correlation

Name: _____

1) A student wonders if tall women tend to date taller men. She measures herself, her dormitory roommate, and the women in the adjoining rooms; then she measures the next man each woman dates. Here are the data (heights in inches):

Women	66	64	66	65	70	65
Men	72	68	70	68	71	65

a) Is there a clear explanatory variable and response variable in this setting? If so, tell which is which. If not, explain why not.

b) Make a well-labeled scatterplot of these data.

c) Based on the scatterplot, describe the pattern, if any, in the relationship between the heights of women and the heights of the men they date.

d) Suppose another 70-inch-tall female who dated a 73-in-tall male were added to the data set. How would this influence r ?

_____ 2) From tax records, it is relative easy to determine the amount of liquor consumed per capita and the number of cigarettes consumed per capita for each of the 10 provinces of Canada. These are plotted on a scatterplot and a high positive correlation is found. Which of the following is correct?

- (A) This implies that heavy smoking causes people to drink more.
- (B) This implies that heavy drinking causes people to smoke more.
- (C) We cannot conclude cause and effect, but this also implies that there is a high positive correlation between cigarette smoking and alcohol consumption for individuals.
- (D) This could be an example of a correlation caused by a common cause because both activities are highly correlated with average family income and average income varies widely among the provinces.
- (E) We cannot conclude cause and effect, but this also implies that the same individuals both smoke and consume liquor.

_____ 3) Suppose a study finds that the correlation coefficient relating family income to SAT scores is $r = +1$. Which of the following are proper conclusions?

- I. Poverty causes low SAT scores.
- II. Wealth causes high SAT scores.
- III. There is very strong association between family income and SAT scores.

- (A) I only
- (B) II only
- (C) III only
- (D) I and II
- (E) I, II and III

_____ 4) An agricultural economist says that the correlation between corn prices and soybean prices is $r = 0.7$. This means that

- (A) when corn prices are above average, soybean prices also tend to be above average.
- (B) there is almost no relation between corn prices and soybean prices.
- (C) when corn prices are above average, soybean prices tend to be below average.
- (D) when soybean prices go up by 1 dollar, corn prices go up by 70 cents.
- (E) the economist is confused, because correlation makes no sense in this situation.

_____ 5) If data set A of (x, y) data has correlation $r = 0.65$, and a second data set B has correlation $r = -0.65$, then

- (A) the points in A fall closer to a linear pattern than the points in B.
- (B) the points in B fall closer to a linear pattern than the points in A.
- (C) A and B are similar in the extent to which they display a linear pattern.
- (D) you can't tell which data set displays a stronger linear pattern without seeing the scatterplots.
- (E) a mistake has been made— r cannot be negative.